The effect of the concentration of a basic or acidic solution on its conductivity and pH

Intended Learning Outcomes

- Be able to design and build a simple electrical current indicator device
- Be able to carry-out a dilution to a specified concentration
- Be able to calculate a pH and measure it using indicator solution
- Be able to describe the effects of concentration of acids and bases on conductivity and pH

Introduction

In this experiment hydrochloric acid and sodium hydroxide solutions will be diluted. The aim of this experiment is to determine what effect diluting these solutions has on their conductivities and their pH's. The first thing we have to do is to make a current detector.

A simple LED circuit

The current detector consists of a simple circuit used to drive an LED (light emitting diode). The power supply for the circuit will be a 9V battery and the components will include wires connected to a snap-on battery cap, an LED and a resistor.

Why do we need a resistor? The 9V battery typically produces a current of up to a few hundred milliamps (mA), whereas our LED bulb has a maximum rating of 20 mA. If we connected it directly to the battery it may work for a few seconds/minutes but it would become hot and eventually may burn-out (don't try this).

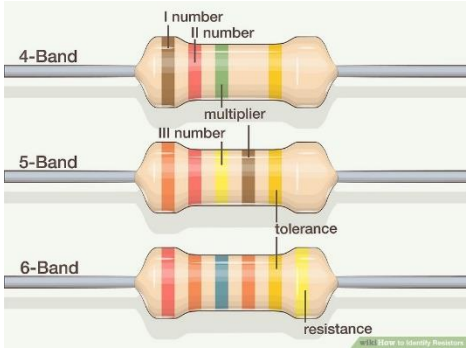
We use a resistor to control the current delivered to the LED. Although the LED has a maximum rating of 20 mA anything between a few to 10 mA provides a perfectly adequate light.

We need to know what resistor to use in our circuit. We can use Ohm's Law to calculate the lowest gain resistor to limit the current to 20 mA.

Voltage of the battery, Vb = 9V
Voltage drop across the LED (obtained from the technical specification sheet) Vf = 4V
I = maximum current of the LED (20mA or 0.02 A).

Use this form of Ohm's Law to find the resistance.

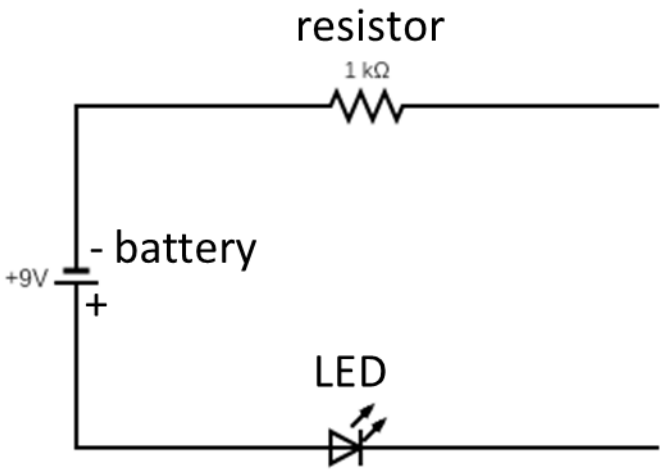
Resistance (ohms) = (Vb – Vf)/I = …250….. ohms



We can use a higher gain resistor than the value calculated and it will still provide enough current for the LED (remember it only needs a few mA).

Resistors are colour coded, usually with 4-6 colour bands. The trick is to know which direction to read them, usually there is a notable wider gap between the final two bands, although it is not always easy to spot on a 6-band resistor.

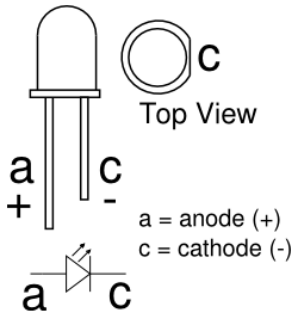
There are a couple of different types of resistor available for your circuit, see if you can work out the resistance of the one that you have – using an online tool is easiest.
https://uk.farnell.com/resistor-colour-code-calculator
The resistors are either 640 (4-band - blue-yellow-black-black) or 1000 (3-band - brown-black-red) ohm resistors, so the currents are 7.8 or 5 mA respectively.



Having determined that your resistor is suitable you can build your circuit. It is always a good idea to draw the circuit first so you know how to connect components together and avoid accidently damaging them.

Notice in the diagram left, that I have not closed the circuit – the two wires will act as electrodes. I can place them in a solution to see if it is conducting. Also, it does not matter where you put the resistor in the circuit (or which way around you connect it).

However the connections to the LED are important (diodes have polarity, see the diagram right). Current flows from the anode (positive, long wire) to the cathode (negative, short wire), so ensure that the anode is connected to the red (+) wire from the battery. It does not matter if you make a mistake, you won't damage the LED, but it won't light. Once assembled, check your circuit is working by touching the two ends of the wires. The LED should light-up.



Now we can move to the experiment

Experimental Procedure

Apparatus
2 x 10 ml beakers
1 x 10 ml syringe
1 x current indicator with connections

Chemicals
Sodium hydroxide NaOH(aq) (0.10 M)
Hydrochloric acid HCl(aq) (0.10 M)
Universal indicator test paper (for pH)
De-ionised water

Procedure

1. Draw up a few millilitres (ml) of **water** into your syringe. Then using a beaker to collect the drops, carefully depress the syringe counting the number of drops equivalent to a volume of 0.5 ml. Repeat 3 times to check for repeatability. Calculate the volume per drop as follows:

   0.5/No.drops = …0.04……..ml/drop
   (based on my average of 12.5 drops = 0.5 ml, this may vary a little)

2. Use the syringe to add 0.1 ml of sodium hydroxide (0.10 M) into Beaker 1.
3. Rinse the syringe with water to clean it.
4. Add 1.9 ml of water into Beaker 1.
5. Swirl the solution in Beaker 1 to mix the contents.
6. Use the syringe to take-up 0.1 ml of the solution in Beaker 1 with the syringe and place this into Beaker 2.
7. Rinse the syringe with water to clean it. Dispense 1.9 ml of water into Beaker 2.
8. Swirl the solution in Beaker 2 to mix the contents.
9. Insert the electrodes of the current meter into the solution in Beaker 1. Observe what happens to the light emitting diode (LED). Record your result in the Table below.
10. Wipe the electrodes clean and then test the conductivity of the solution in beaker 2 in the same way. Record your result in the Table below.
11. Test the pH in beakers 1 and 2 using the universal indicator paper strips. Record your result in the Table below.
12. Rinse and dry the beakers and syringe and repeat steps 1 to 10 as before, using hydrochloric acid (0.10 M) instead of sodium hydroxide.

Table: Experimental Results

Beaker	Concn. NaOH (M)	LED – dull/bright	pH of solution
1	0.005	bright	12
2	0.00025	dull	10
Beaker	Concn. HCl (M)	LED – dull/bright	pH of solution
1	0.005	bright	2
2	0.00025	dull	4

Questions

1. Calculate the concentration of each of the sodium hydroxide and hydrochloric acid solutions. Write these down in Table 1.
   Same for both HCl and NaOH
   $M_1V_1 = M_2V_2$
   Beaker 1: $0.1 \times 0.1 = M_2 \times 2 = 0.005$ M
   Beaker 2 = 0.00025 M

2. Calculate the predicted pH of each of the sodium hydroxide and hydrochloric acid solutions. How do these compare with your measured values?
   NaOH
   Beaker 1:
   [OH-] = NaOH = 0.005
   pOH = -log[OH-] = -log 0.005 = 2.3
   We are finding pH so:
   pOH + pH = 14

pH = 14 – 2.3 = 11.7

Beaker 2: pH = 10.4

HCl
Beaker 1:
[H+] = [HCl] = 0.005
pH = -log[H+] = -log 0.005 = 2.3
Beaker 2 – pH = 3.6

3. What causes the current indicator to glow, write the dissolution reactions?
   $HCl\ (aq) = H^+\ (aq) + Cl^-\ (aq)$
   $NaOH\ (aq) = Na^+\ (aq) + OH^-\ (aq)$
   In wires, it is electrons that carry the current, but in solutions the current is carried by ions. NaOH and HCl strongly dissociate into ions in solution.

4. What is the effect of the concentration of a basic or acidic solution on its conductivity and pH?
   The electrical conductance depends upon the total concentration of the ions (and hence pH). The most mobile ions are H^+ and OH^-. So the more acidic or basic the solution the greater the electrical conductivity will be.